

System Architecture

AI-Powered Content Operations Platform — Skip Studio

Prepared for	Skip Studio (fictional client — Exology Pioneer Program, Week 4 Final Project)
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Overview

The parts of the product, and how a campaign moves through them — from intake to a scheduled, dual-approved plan that actually publishes. Two things hold everywhere in this architecture, not just in the component that first checks them: **nothing is scheduled or published without both internal and client approval currently recorded**, and **nothing is drafted that breaks a rule**— both enforced in application code, never in a prompt.

An account manager submits a campaign brief — or converts a client's own calendar request — for a client operating in a single market (Saudi Arabia, to start). The system looks up that client's rules, resolves upcoming occasions relevant to their market, and drafts a full multi-form content plan: posts, images, videos, reels, and photoshoot briefs, each grounded in the specific rule it was written under. An assigned creator refines it, an internal reviewer approves it, the client gives final sign-off, and only then is anything scheduled — and only then does a scheduler actually publish it to the client's connected Instagram account. Either the reviewer or the client can pull an item back even after approving it, right up until it goes live. Performance metrics flow back in once a post is published, visible to the account manager and the client.

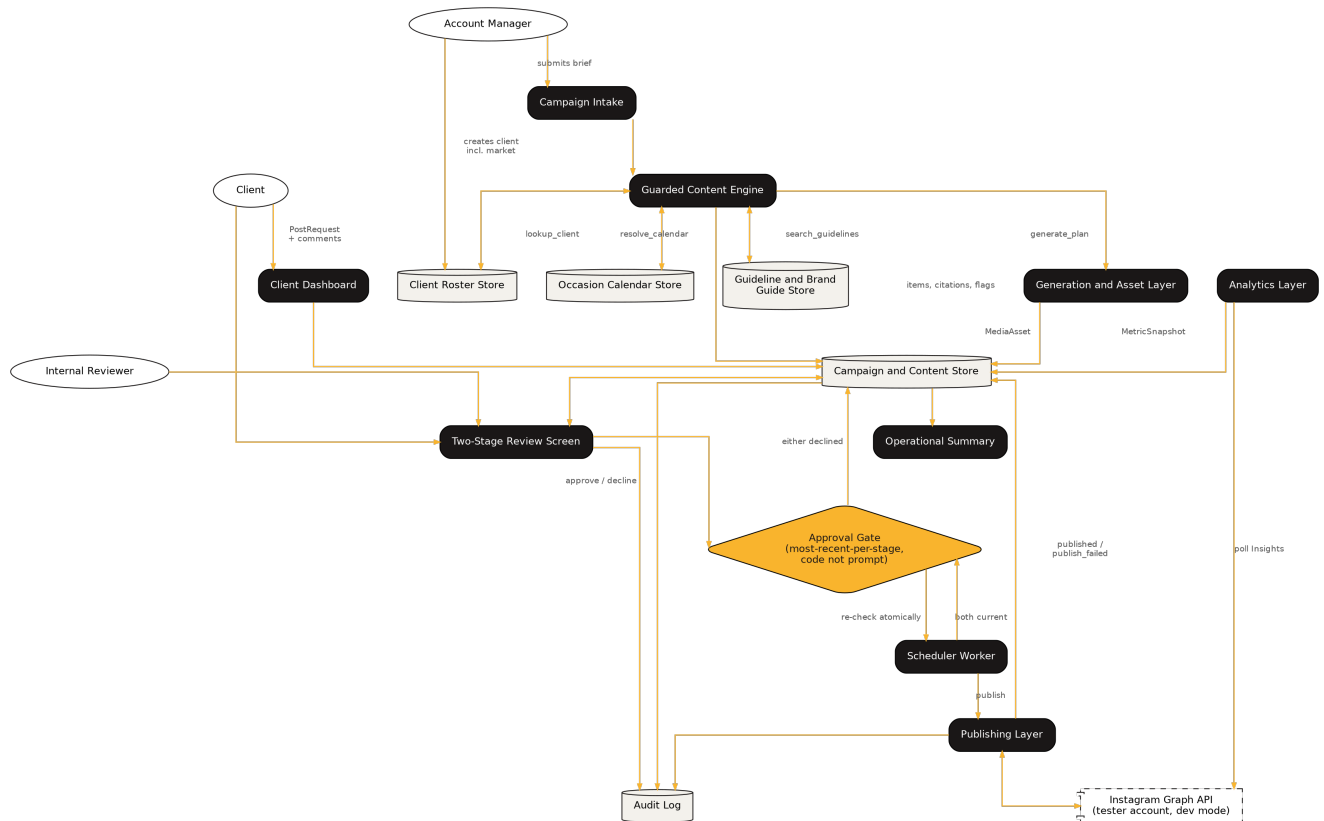
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Components

COMPONENT	ROLE
Campaign Intake	Accepts a brief through a form, an incoming queue, or a converted client PostRequest; normalizes it into a structured object the engine can use.
Guarded Content Engine	The guarded pipeline that turns a brief into a drafted, cited, multi-form plan — or a request, a flag, or a refusal. Also exposes a narrower, item-level regeneration path a content creator can prompt directly, with the same grounding discipline.
Client Roster Store	The client roster: status, channels, market, governing brand guide, account manager. Account managers create clients directly here.
Guideline & Brand Guide Store	Agency standards plus each client's versioned brand guide, retrieved per-request — never held in the model's memory.
Occasion Calendar Store	Seeded regional and local occasions, scoped by market, Hijri-aware for dates that move year to year.
Campaign & Content Store	Transactional data: campaigns, content items, media assets, citations, approvals, flags, platform connections, metrics, post requests, comments, audit log.
Generation & Asset Layer	Image and video generation clients. Photoshoot briefs are planning artifacts — shot lists and schedule slots — not generated assets.
Publishing Layer	The Instagram Graph API client, the OAuth connect flow, and a scheduler worker that re-checks the approval gate atomically before every publish attempt.
Analytics Layer	Polls Instagram Insights for published items on a fixed schedule and writes time-series metric snapshots.
Scheduling / Background Jobs	Time-triggered publish attempts, periodic analytics polling, periodic occasion refresh.
Two-Stage Review Screen	Where internal reviewers and clients act on drafted items; the only place approvals — and declines, including after the fact — are recorded.
Client Dashboard	Where a client sees their assigned account manager, requests or reschedules a post on the calendar, and comments on requests or existing posts.
Brand Guide Management	Where an account manager edits a client's guide; versions and gates changes behind client approval.
Operational Summary	Counts panel: every client's pipeline by status, market, and upcoming occasion.
Audit Log	Append-only record of edits, schedules, reschedules, deletes, publish attempts, take-downs, and late-revokes.

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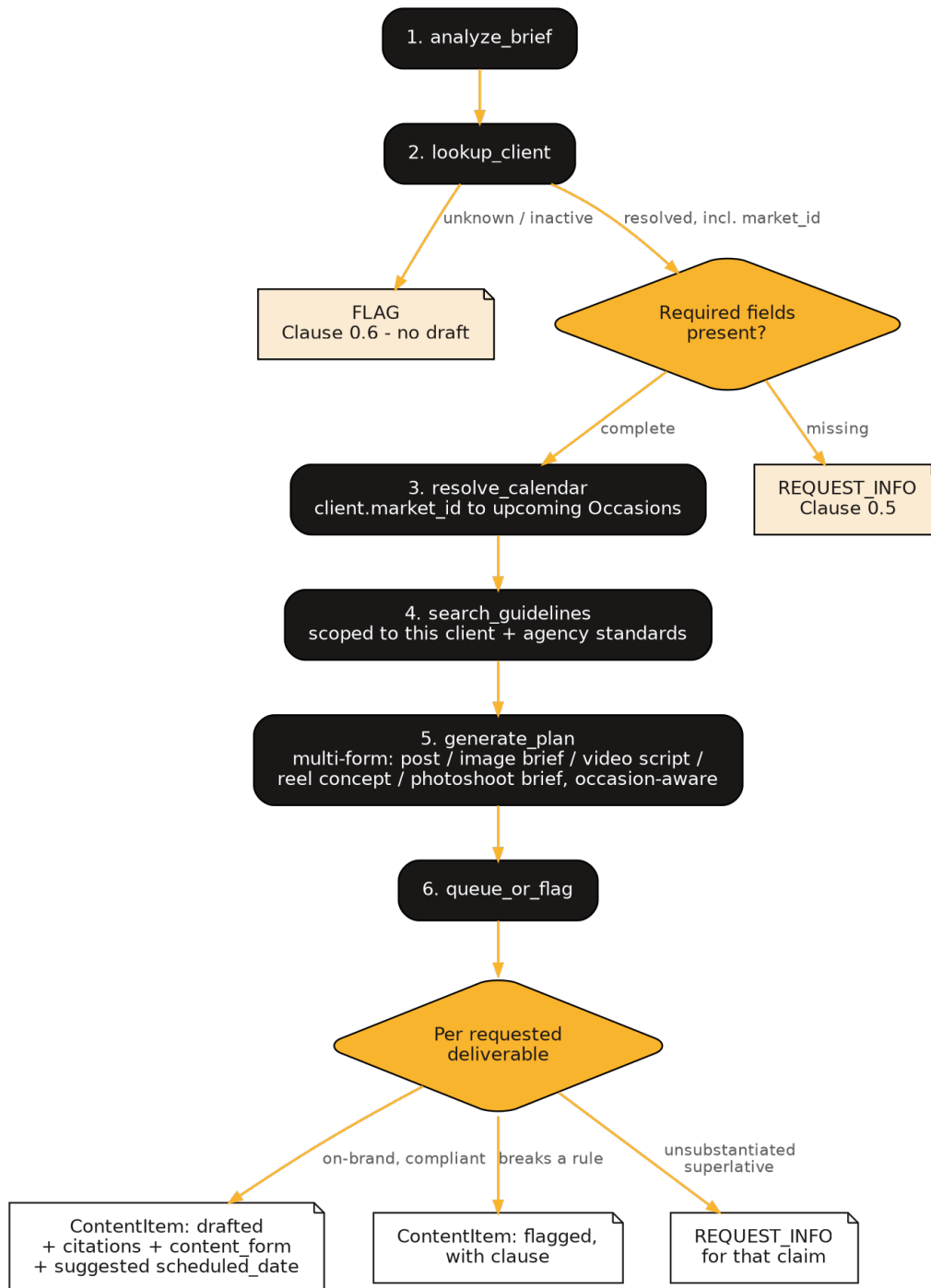
Component diagram



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Guarded Content Engine

For every brief, the engine runs a fixed sequence — retrieval and rule-checking are deterministic code, not model calls; only extraction and drafting are.



`resolve_calendar` pulls `Occasion` rows for the client's `market_id` within the campaign's planning window, resolving Hijri-based occasions (Ramadan, Eid al-Fitr, Eid al-Adha) against the seeded `OccasionDate` table rather than computing them live. `generate_plan` receives this as context so its proposed `scheduled_date`s and content angles can reference real upcoming dates — but the engine only proposes; nothing is actually scheduled without the same approval gate as everything else.

`search_guidelines` is retrieval-scoped: a request can only return the requesting client's own active brand guide version plus the agency standards handbook. It is architecturally impossible for one client's rules — or content — to leak into another client's retrieval or drafting.

Trends are treated as an agency-curated internal signal — a field the account manager or content lead can attach to a brief, not a live external trends API. Instagram's own Graph API doesn't expose a general trends or trending-audio endpoint to third-party apps.

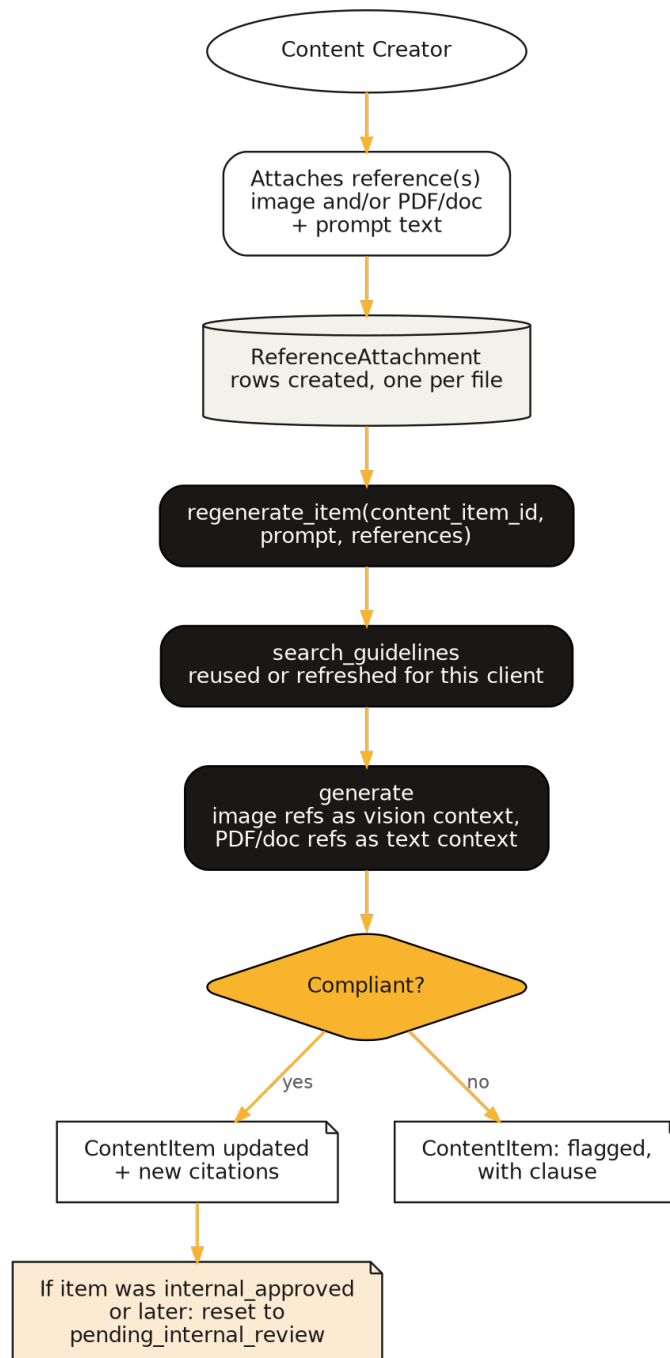
Photoshoots are treated as a planning artifact — a shot list, mood board, and schedule slot. No API generates a real-world physical shoot; `generate_plan` produces the brief, a human executes it, and the resulting media is uploaded afterward as a `MediaAsset` like any other.

Sensitive-sector clients (healthcare, financial, government) get every resulting item tagged `compliance_review_required` at this step, even when every item is otherwise clean.

Item-level regeneration

Once an item exists, a content creator can prompt the engine again on that one item directly — attaching a reference image and/or a PDF/doc, with

instructional text — rather than re-running the whole brief. This is a narrower operation than intake: it skips `analyze_brief`, `lookup_client`, and `resolve_calendar`, since those are already resolved, but still runs `search_guidelines` and still checks the result against the client's rules before accepting it.



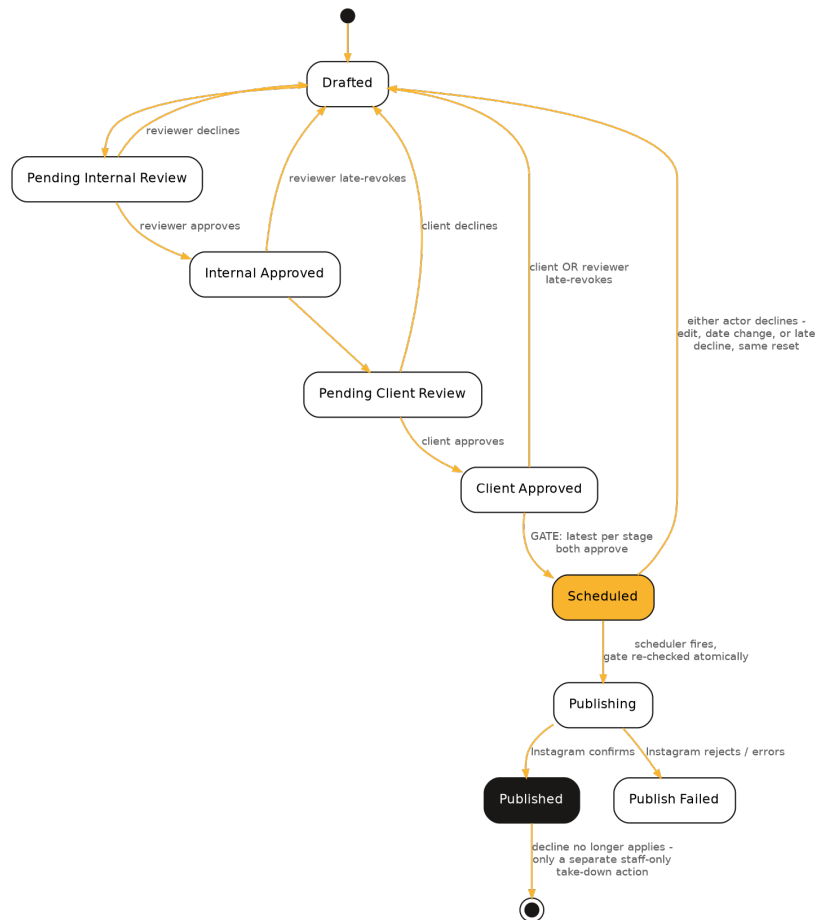
An image reference is passed as vision context — it shapes how the result looks. A PDF or doc reference is text-extracted and passed as context — it shapes what the result says or knows. Only a content creator can attach references; each attachment belongs to the one `ContentItem` it was supplied for, and rows accumulate across regenerations rather than being overwritten, so a reviewer can see exactly which reference produced which version.

A reference file carries no authority over compliance, the same way brief wording and client comments never do — a competitor screenshot attached as "make it look like this" still can't produce disparaging content, and a reference photo showing an unapproved claim doesn't make that claim draftable. A regeneration on an item that has already reached `internal_approved` or later is a content edit like any other, and resets it to `pending_internal_review` under the same rule as every other invalidation.

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Approval gate

The gate answers one question: **is the most recent recorded decision for each stage — internal and client — currently an approval**. It is called every time a schedule or publish is attempted, never assumed from a prior state or a brief's wording.



Decline is available at `pending_internal_review` / `pending_client_review` in the normal course of review, and remains available afterward — at `internal_approved`, `client_approved`, and `scheduled` — from either the internal reviewer/content lead or the client, so either party can pull back something they already signed off on. Any invalidation, whichever party or cause triggers it — a content edit, a `scheduled_date` change, or a late decline — produces the identical reset to `pending_internal_review`, unscheduling if scheduled. One rule, no per-cause branching.

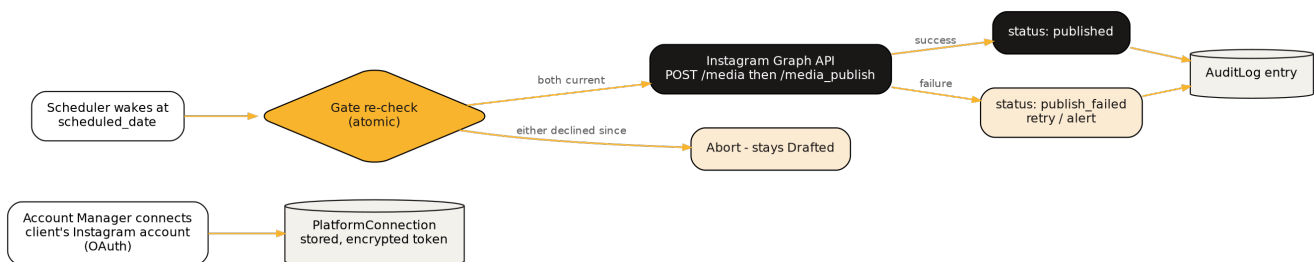
A client `Comment` never triggers this. `Comment` is a discussion thread on a `PostRequest` or a `ContentItem`; only a formal `Approval` row with `decision = decline`, or a deliberate content/date edit by staff, invalidates approvals.

The scheduler's gate re-check is atomic — check-then-publish as one guarded step, not two separate calls — so a decline landing a second before the scheduled time cannot lose a race against the publish call. This is a dedicated test case: fire a decline concurrently with a scheduled publish attempt and assert the publish never happens.

Once `published`, decline no longer applies. The only remaining lever is a separate, staff-only **take-down** action that calls Instagram's delete endpoint directly — kept out of the approval vocabulary on purpose, since "declined" and "removed from a live account" are different classes of action with different risk.

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Publishing flow



Meta requires every Instagram account touched by the API to be a Professional account (Business or Creator) — personal and private accounts have no API path at all, and Professional accounts cannot themselves be private. Full production access, publishing to arbitrary client accounts, requires Meta App Review, a 2–4 week process. `PlatformConnection` is built and tested against one tester-added Professional account in Meta's Development mode, which requires no App Review — a real, working integration, honestly scoped to one connected account rather than any client's account. Extending to arbitrary client accounts later is an App Review submission, not a code change.

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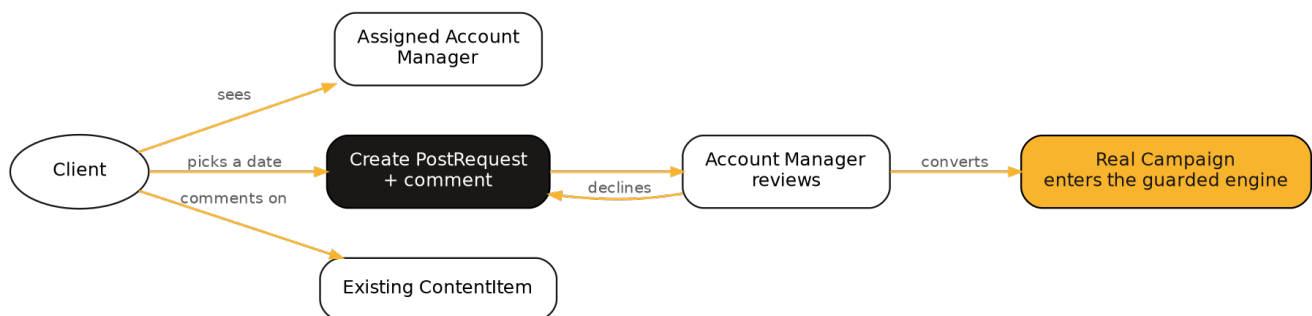
Analytics flow

A separate scheduled job — not part of the request/response path — polls Instagram Insights for every `ContentItem` in `published` status on a fixed interval, writing one `MetricSnapshot` row per metric per poll: a time series, not an overwritten current value. Aggregation queries roll item-level snapshots up to campaign level.

Account Manager (their assigned clients only) and Client (their own data only) can view analytics. Agency Admin retains its cross-client view. Content Creator access to analytics is not granted.

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Client Dashboard flow



`PostRequest` carries no authority, same as brief wording never does — however detailed the client's comment, it is a front door into the same pipeline, not a bypass. It cannot become a scheduled item without an account manager deliberately converting it into a real `Campaign` and the full engine and gate running as normal.

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Roles and access

ROLE	CAPABILITIES
Account Manager	Creates clients directly, including market; submits briefs; is the internal reviewer by default; connects and manages a client's <code>PlatformConnection</code> ; converts a <code>PostRequest</code> into a real <code>Campaign</code> ; can late-revoke on any of their clients' approved or scheduled items.
Content Creator (includes <i>Marketing Specialist</i>)	Authors the full multi-form plan — post, image, video, reel, and photoshoot briefs; has calendar and occasion visibility; refines drafts before internal review; attaches image or PDF/doc reference material when prompting generation or regeneration of a specific item. Does not trigger publish directly — staging only; scheduling stays gate-controlled regardless of who staged the work.
Content Lead	When assigned in place of the account manager, acts as the internal reviewer for that client, with the same late-revoke power.
Client	Sees their assigned account manager; requests or reschedules posts and comments through the dashboard; gives final approval on content and on brand guide changes; can late-revoke their own approval on an approved or scheduled item; views analytics on their own data only.
Agency Admin	Assigns each client's account manager, reviewer, and creators; a cross-client view of where every account stands; not involved in day-to-day content work.

`PlatformConnection` credentials are staff-only, never exposed to a client contact — a client can see that their account is connected, never the token itself.

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Design decisions

DECISION	CHOICE	WHY
Market	A real table, single-select per client, seeded with one row (Saudi Arabia)	A second market later is a data insert, not a migration; no sub-national region is modeled for now
Hijri-calendar occasions	Seeded per-year lookup table	No external calendar-conversion dependency; documented as needing yearly reseeding
Trends	Agency-curated internal signal	No accessible third-party trends endpoint on Instagram's API
Photoshoots	Planning artifact, not a generated asset	No API produces a real-world shoot

Instagram scope	One tester-added Professional account, Development mode	Full App Review takes longer than the build window
Gate semantics	Most recent Approval per stage, not any-approval-ever	Required for a late decline to actually take effect
Late-revoke	Symmetric for reviewer and client, same statuses, same reset	One code path; reviewer accountability matters as much as client sign-off
Comment vs. Approval	Comments never trigger a reset; only a formal decline or an edit does	Keeps the gate from being eroded by informal wording
Publish/decline boundary	Decline valid through scheduled; not valid once publishing or published	A live post needs a take-down action, not a retroactive decline
Publish trigger	Gate and scheduler only, never a manual creator action	Keeps the broadened creator role from becoming a bypass of the gate
Analytics access	Account Manager and Client only	Content Creator access not required for this scope
Retrieval scope	search_guidelines hard-scoped to one client plus agency standards	Makes cross-client leakage structurally impossible rather than a matter of model behavior
Reference attachments	Creator-only, scoped to one ContentItem's prompt; image and PDF/doc only, no video	Keeps generation-time reference material distinct from compliance grounding; video-conditioned generation is a materially bigger integration for this build window